

E076

Applied Software Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A unit test for OrderService's tax calculation calls the real payment gateway, so it is slow and fails whenever the gateway is down. The team's CI runners were upgraded last month. How should the test be fixed?

- A. Run the test only at night when the gateway is quieter.
- B. Upgrade the CI runners again.
- C. Delete the test.
- D. Replace the gateway with a mock or stub in the unit test.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A test starts the real API and a real database container, sends a request, and checks that a record is saved. What kind of test is this?

- A. An integration test
- B. A load test
- C. A static code analysis
- D. A unit test

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A developer who joined the team in June fixes a function after its unit test fails, reruns only that one test, sees it pass, and merges. The project has 600 tests that take 4 minutes to run. What should the developer have done?

- A. Rerun the full suite to catch regressions, then document the change.
- B. Edit other tests so they match whatever the code now returns.
- C. Wait until a more senior developer joins.
- D. Raise the coverage target to 100%.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A billing module has 100% line coverage and a green build badge on its README. A divide-by-zero crash still reaches production when an invoice has zero items. What does this show?

- A. Remove the build badge.
- B. Coverage shows code ran, not that edge cases were checked; add a test with an assertion for zero items.
- C. The coverage tool is broken.
- D. 100% coverage means the code is bug-free, so the crash must be a hardware fault.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An engineer, working late before a holiday, fixes a config file by opening a shell inside a running container. After the next deployment, the fix is gone. What explains this?

- A. The holiday freeze rolled it back.
- B. The container saved the change back to the image, but the image was corrupted.
- C. New containers are created from the unchanged image; put the fix in the image or supply it through config or a volume.
- D. Containers share a kernel, so the host overwrote the file.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

What does running docker run on an image create?

- A. A new image
- B. A running container from that image
- C. A Dockerfile
- D. A virtual machine with its own kernel

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A Python service's Dockerfile, maintained by a team of three, runs `COPY . .` and then `RUN pip install -r requirements.txt`. Every one-line code change triggers a 6-minute rebuild. What change helps most?

- A. Switch to a larger base image.
- B. Build with the cache turned off.
- C. Split the team so fewer people change code.
- D. Copy requirements.txt and install dependencies first, then copy the rest of the source.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A web app's database address is hard-coded in the image. Staging and production need different addresses, and production uses a password. The app has been live for 18 months. What is the best practice?

- A. Build one image and inject settings through environment variables, with the password from a secrets manager.
- B. Rewrite the app since it is 18 months old.
- C. Build a separate image for each environment with its values hard-coded.
- D. Commit the production password to the repository so deploys are simple.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A streaming company with 8 million subscribers wants to release a new version to about 5% of users, watch error rates for an hour, and then expand if all is well. Its office is in Seattle. Which strategy fits?

- A. Recreate: stop the old version, then start the new one
- B. Blue-green deployment
- C. Canary deployment
- D. Deploy only to the Seattle office

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A team keeps two identical production environments and wants to move all traffic to the new version at once, with the ability to switch everything back instantly. Which strategy is this?

- A.** Canary deployment
- B.** Rolling deployment
- C.** Blue-green deployment
- D.** Hotfix

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A public API library at version 2.4.1, downloaded 30,000 times a week, removes an endpoint that clients use. What should the next version be?

- A.** 3.4.1
- B.** 2.5.0
- C.** 2.4.2
- D.** 3.0.0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A library at 1.9.3 adds a new optional parameter; existing calls still work. What is the next version?

- A.** 2.0.0
- B.** 1.10.0
- C.** 1.9.4
- D.** 1.10.3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A unit test for OrderService's tax calculation calls the real payment gateway, so it is slow and fails whenever the gateway is down. ~~The team's CI runners were upgraded last month.~~ How should the test be fixed?

Target word: isolate a unit **Decides:** The tax logic does not need the real gateway.

A	Adjacent category	Still depends on the real gateway.
B	Camouflage	Faster machines do not remove the dependency.
C	Dropped element	Loses coverage of the tax logic.
D	Correct	Isolates the logic under test; fast and reliable.

2. Answer A Concept 1 · transfer

A test starts the real API and a real database container, sends a request, and checks that a record is saved. What kind of test is this?

Target word: classify the test **Decides:** Real components working together.

A	Correct	Checks the API and database together.
B	Adjacent category	Load tests measure behavior under heavy traffic.
C	Confusion set	Analysis does not run the code.
D	Inverted operation	Units are tested in isolation with dependencies mocked.

3. Answer A Concept 2

A developer ~~who joined the team in June~~ fixes a function after its unit test fails, reruns only that one test, sees it pass, and merges. The project has 600 tests that take 4 minutes to run. What should the developer have done?

Target word: verify a fix **Decides:** A change can break other code.

A	Correct	Four minutes is a small cost to catch side effects.
B	Inverted operation	Changes tests to fit the code instead of checking the code.
C	Camouflage	Tenure does not replace running the suite.
D	Adjacent category	Does not check this change for regressions.

4. Answer B Concept 3

A billing module has 100% line coverage ~~and a green build badge on its README.~~ A divide-by-zero crash still reaches production when an invoice has zero items. What does this show?

Target word: limits of coverage **Decides:** The lines ran, but the zero case was never checked.

A	Camouflage	The badge is not the issue.
B	Correct	Behavior needs targeted assertions.
C	Confusion set	It reported correctly that lines ran.
D	Absolute language	Coverage does not prove correctness.

5. Answer C Concept 4

An engineer, ~~working late before a holiday~~, fixes a config file by opening a shell inside a running container. After the next deployment, the fix is gone. What explains this?

Target word: container lifecycle **Decides:** Deployment creates new containers from the image.

A	Camouflage	Timing is not the cause.
B	Inverted operation	Containers do not write back to their image.
C	Correct	Container changes are not saved to the image.
D	Confusion set	Kernel sharing does not reset files.

6. Answer B Concept 4 • transfer

What does running docker run on an image create?

Target word: image-to-container direction **Decides:** An image is instantiated as a container.

A	Inverted operation	Images are built, not created by running.
B	Correct	Containers are instances of images.
C	Adjacent category	The Dockerfile is the recipe used to build an image.
D	Confusion set	Containers share the host kernel.

7. Answer D Concept 5

A Python service's Dockerfile, ~~maintained by a team of three~~, runs COPY . . and then RUN pip install -r requirements.txt. Every one-line code change triggers a 6-minute rebuild. What change helps most?

Target word: use layer caching **Decides:** Copying source first invalidates the install layer on every edit.

A	Adjacent category	Size does not fix step order.
B	Inverted operation	Makes every build slower.
C	Camouflage	Team size is not the cause.
D	Correct	The install layer stays cached until dependencies change.

8. Answer A Concept 6

A web app's database address is hard-coded in the image. Staging and production need different addresses, and production uses a password. ~~The app has been live for 18 months~~. What is the best practice?

Target word: config per environment **Decides:** One image, settings supplied at runtime.

A	Correct	Same artifact everywhere; config and secrets stay outside.
B	Camouflage	Age does not decide config handling.
C	Adjacent category	Different artifacts per environment and secrets baked in.
D	Dropped element	Drops secret protection entirely.

9. Answer C Concept 7

A streaming company ~~with 8 million subscribers~~ wants to release a new version to about 5% of users, watch error rates for an hour, and then expand if all is well. ~~Its office is in Seattle.~~ Which strategy fits?

Target word: choose deployment strategy **Decides:** A small share of traffic first.

A	Inverted operation	All users at once with downtime.
B	Confusion set	Switches all traffic at once between two environments.
C	Correct	Gradual exposure by traffic share.
D	Camouflage	Office location is not a traffic strategy.

10. Answer C Concept 7 • transfer

A team keeps two identical production environments and wants to move all traffic to the new version at once, with the ability to switch everything back instantly. Which strategy is this?

Target word: name the strategy **Decides:** Two full environments; one switch.

A	Inverted operation	Gradual, not all at once.
B	Adjacent category	Replaces instances in batches in one environment.
C	Correct	Flip traffic between two environments.
D	Confusion set	A type of change, not a deployment strategy.

11. Answer D Concept 8

A public API library at version 2.4.1, ~~downloaded 30,000 times a week,~~ removes an endpoint that clients use. What should the next version be?

Target word: version for a breaking change **Decides:** Removing an endpoint breaks clients.

A	Dropped element	Forgets to reset MINOR and PATCH.
B	Adjacent category	MINOR is for backward-compatible features.
C	Confusion set	PATCH is for compatible fixes.
D	Correct	Breaking change: bump MAJOR and reset the rest.

12. Answer B Concept 8 • transfer

A library at 1.9.3 adds a new optional parameter; existing calls still work. What is the next version?

Target word: version for a compatible feature **Decides:** Backward-compatible new feature.

A	Inverted operation	MAJOR is for breaking changes.
B	Correct	MINOR goes up; PATCH resets.
C	Confusion set	PATCH is for fixes, not features.
D	Dropped element	Forgets to reset PATCH.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Unit vs. integration tests and mocks	Unit tests check one unit in isolation, replacing outside dependencies with mocks or stubs; integration tests check real components working together.	Calling a test that hits a real database or API a unit test.	Whether the test touches real external services.
2	Fix, then rerun the whole suite	After changing code to fix a failing test, rerun the full test suite to catch regressions, then record the change.	Rerunning only the one failing test, or editing the test to match wrong output.	A fix shipped after checking a single test.
3	Coverage is not correctness	Coverage shows which code ran during tests, not whether behavior was checked; edge cases and meaningful assertions matter.	Treating 100% coverage as proof of no bugs.	High coverage and a bug in production anyway.
4	Image vs. container	An image is a read-only template; a container is a running instance created from it. Changes made inside a container vanish when it is replaced unless stored in a volume or baked into the image.	Patching a running container and expecting the change to survive a redeploy.	A fix made “inside the container” that disappears.
5	Dockerfile layer caching	Order steps from least to most frequently changing: copy the dependency list and install dependencies before copying source code.	Copying all source first, so every code edit reinstalls every dependency.	Slow rebuilds after tiny code changes.
6	Config and secrets outside the image	Build one image and supply environment-specific settings through environment variables or a secrets manager at runtime.	Building a separate image per environment with hard-coded settings.	Different settings for dev, staging, and production.
7	Deployment strategies	Canary sends a small share of traffic to the new version first; blue-green switches all traffic between two full environments for instant rollback; rolling replaces instances in batches.	Mixing up canary (gradual, by traffic share) and blue-green (all at once, two environments).	“A small percentage of users” vs. “switch everything back instantly.”
8	Semantic versioning	MAJOR.MINOR.PATCH: bump MAJOR for breaking changes, MINOR for backward-compatible features, PATCH for backward-compatible fixes; reset the lower numbers when a higher one goes up.	Bumping MINOR for a breaking change, or forgetting to reset lower numbers.	A version number and a described change.

Course scope (public description): Applied Software Engineering demonstrates full stack software solutions by producing comprehensive documentation and building functional applications that integrate front end, back end,

and supporting services. It executes unit testing, revises software based on test outcomes, and documents all updates to ensure accuracy and quality. The course also deploys software applications to a cloud platform, using scripts and containers to deliver consistent, reliable performance across environments.